

[Extended Abstract] Towards Golden Standards for Quantifying Privacy of Synthetic Tabular Data

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Abstract—The rise of Generative Artificial Intelligence (GenAI) techniques has created numerous opportunities but also a few challenges, the most prominent of which relates to privacy and data confidentiality. Many different privacy metrics, evaluation techniques (and variants thereof) have been proposed and used in literature to evaluate the privacy of synthetic tabular data. These are roughly classified as (i) non-adversarial metrics – evaluating and comparing the data statistically–, and (ii) adversarial metrics –approaches that simulate adversarial approaches (privacy attacks) and evaluate their success. There is however a large discrepancy between these metrics. The adversarial metrics require significant resources (CPU, time, memory) whereas the non-adversarial metrics are less costly to calculate. In this paper, we highlight the lack of a ‘golden standard’ for privacy auditing, i.e. for evaluating and quantifying the privacy of synthetic tabular data. We specifically present a research framework to support extensive experimentation with the intent to obtain the necessary knowledge and better inform metric selection and enable pragmatic trade-offs.

Index Terms—Synthetic tabular data, Privacy metrics, Privacy evaluation

I. INTRODUCTION

Synthetic tabular data generation offers a promising solution for preserving privacy and confidentiality [1]. In this approach, original datasets are replaced with their synthesized counterparts which do not consist of personal data, and thus theoretically preserve the privacy and confidentiality of the original data records. In contrast to more traditional privacy preservation techniques such as the tailored de-identification of datasets [2], synthetic data generation can be extensively automated; relies significantly less on manual experimentation and trial-and-error; can achieve similar utility results [3], [4]; does not impose any limits on data volume; and can even lead to overall increases in data quality [5]–[7].

However, the use of synthetic tabular datasets is not without any residual privacy risks. There is the risk of overfitting to the original data set, which leads to the generation of records that are overly specific to individual records of the original data set [8]. Additionally, concrete attacks have emerged –and shown effective– against synthetic tabular datasets, such as the inference and linkage for example of membership in the original datasets [8], or of attributes of the original datasets [9].

The quantitative privacy assessment of a synthetic tabular dataset is an active area of research. Seminal studies on synthetic tabular data generators rely on statistical approaches to measure privacy risks [10]–[12]. Their core theoretical

foundation relies on statistical methods to measure similarity, under the assumption that a higher degree of similarity between synthetic and real data correlates with increased privacy leakage. Although these statistical approaches are theoretically and computationally not complex (and thus easy to apply), they fail to account for the complexities of real-world privacy threats [13]. On the other hand, the adversarial machine learning community focuses more extensively on attack-based privacy assessments, also known as *adversarial metrics*, which provide potentially more robust evaluations. However, their effectiveness varies significantly depending on the assumptions made about the attacker, such as prior knowledge, capabilities, and their access to auxiliary data that may share common attributes (columns) or entries (rows) with the training data of the generative model. In addition, adversarial metrics are computationally expensive because of the need to evaluate and enumerate different attack models, and conduct the attacks on a per-record basis. This complexity quickly renders a fully exhaustive assessment infeasible. For example, as noted by [14], assessing synthetic tabular datasets using state-of-the-art membership inference attacks (MIA) incurs significant computational costs: even for modest datasets consisting of just 1,000 records, fully estimating the privacy risks could require up to 250 days of compute time [14].

Problem. From the perspective of a privacy auditor that aims to conduct quantitative privacy assessments before releasing a generated synthetic tabular dataset¹, it is neither feasible to meticulously apply all of the available privacy metrics nor straightforward to make informed decisions regarding protection strengths, in part due to the large space of quantitative evaluation approaches available [15] and their heterogeneity and complexity.

Contribution. This extended abstract highlights a knowledge gap in the quantified privacy assessment of synthetic tabular data, and outlines our current research approach towards addressing this gap.

Outline. Sect. II presents the proposed research framework. Sect. III describes the methodology. Finally, Sect. IV summarizes the potential contributions.

II. RESEARCH FRAMEWORK

Fig. 1 presents the overall research framework. The four main research objectives (RO) their and motivations can be

¹Or a trained generative model altogether.

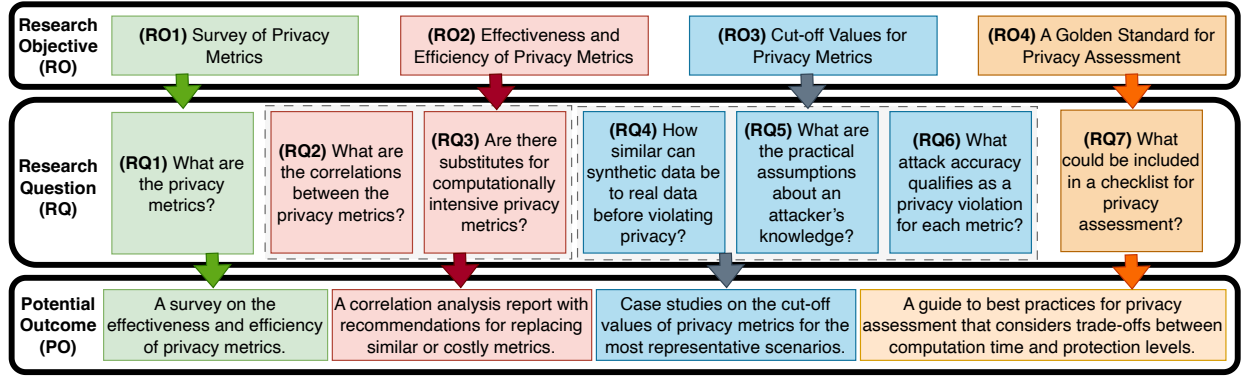


Fig. 1. The overall research framework

further explained as follows:

- **RO1. Survey of Privacy Metrics.** Different privacy metrics used in both literature and existing tools would be surveyed, followed by experimental evaluations of their effectiveness and efficiency.
- **RO2. Correlation Analysis.** The second RO includes analyzing correlations to eliminate redundancy or merge similar metrics. Additionally, we would explore the possibility of replacing computationally expensive metrics with more efficient alternatives while ensuring sufficient coverage of the different privacy notions. The goal is to establish a privacy evaluation framework that is both comprehensive and computationally feasible.
- **RO3. Quantitative Criteria of Privacy Metrics.** We would empirically investigate whether universal cut-off values exist for privacy metrics or if they are scenario-dependent. If privacy thresholds are context-specific, a systematic approach should be established for defining appropriate cut-off values based on quantitative results and expert opinions. Furthermore, methodologies should be defined that allow privacy auditors to confidently determine whether a synthetic tabular dataset meets an acceptable level of privacy protection.
- **RO4. Golden Standards for Privacy Assessments.** The end objective is to determine best practices (golden standards) for privacy assessments of synthetic tabular data with practical assumptions about the attacker's knowledge, providing clear guidance on the essential metrics and evaluation methodologies.

III. DEVELOPMENT OF AN EXPERIMENTAL FRAMEWORK

The four research objectives presented in Fig. 1 can be achieved by establishing an experimental framework, as depicted in Fig. 2. We envision an automated and integrated tool chain that incorporates a reusable metric suite, and supports the execution and workflow of specific privacy evaluation experiments, against a range of variables, such as different threat models, using different datasets, etc.

Building blocks. We have identified several existing open-source implementations of privacy metrics. For non-adversarial

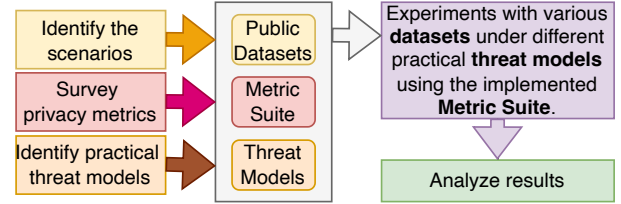


Fig. 2. Experiment Framework

metrics such as the Hitting Ratio, Nearest Neighbor Distance, Close Value Ratio, Similarity Ratio, and Nearest Neighbor Accuracy, implementations can be found in Synthcity [16]. The implementations of Data Likelihood and Detection Rate are available in SDMetrics [17]. Additionally, Anonymeter [18] provides implementations for Singling Out Attacks and Record Linkage Attacks. Finally, both Attribute Inference and Membership Inference Attacks (DOMIAS) have been implemented using Synthcity [16].

IV. CONCLUSION

Quantitative privacy assessment should not be reduced to an uninformed ‘tick-the-boxes-compliance’ effort. Instead, we advocate and strive to support making more informed and well-motivated decisions when it comes to privacy auditing. We present an active research roadmap which includes (i) an effort to survey the different quantitative privacy evaluation approaches and metrics used in literature, (ii) a categorization of these metrics based on the different privacy notions they address, and (iii) experimental evaluation efforts to determine the extent to which these privacy metrics correlate with one another while also assessing their computational cost (i.e., evaluation time). Fundamental understanding of the correlations between different metrics and their computational complexity is required to make effective trade-offs involved in privacy auditing. The envisioned knowledge gain allows practitioners and evaluators to make informed decisions on what metrics to prioritize for practical evaluation before further processing, release, or publication of synthetic tabular datasets. Ultimately, this work aims to facilitate the more systematic selection and

exclusion of privacy evaluation methods in synthetic tabular data, contributing to the development of a comprehensive guide for privacy assessment in synthetic tabular data generation. In the longer term, addressing these challenges will help ensure regulatory compliance and contribute to the broader adoption of synthetic tabular data in real-world applications.

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